Learning Outcomes

In this chapter, you'll learn…

- how to relate the pressure, volume, and temperature of a gas.
- how the pressure and temperature of a gas are related to the kinetic energy of its molecules.
- how the heat capacities of a gas reveal whether its molecules are rotating or vibrating.
- how the speeds of molecules are distributed in a gas.
- what determines whether a substance is a gas, a liquid, or a solid.

Introduction (1 of 2)

- How does the speed of the molecules in the air above the frying pan compare with that of the ones in the rest of the kitchen?
- How do the atoms of a gas determine its temperature and pressure?
- We'll see how the microscopic properties of matter determine its macroscopic properties.

Equations of State and the Ideal-Gas Law

- Quantities such as pressure, volume, temperature, and the amount of a substance are **state variables** because they describe the state of the substance.
- The **equation of state** relates the state variables.
- The ideal-gas equation is an equation of state for an ideal gas:

Ideal-gas equation:

$$pV = nRT$$

Gas pressure → p; Gas volume → V; Number of moles of gas → n; Absolute temperature of gas → T; Gas constant → R

- The **molar mass *M*** (**molecular weight**) is the mass per mole. The total mass of n moles is $m_{total} = nM$.
- Video Tutor Solution: Example 18.1

Question 1

0 / 10 pts

A 5-L tank at 37°C contains 24.7×10^{-4} kg of element 2 listed in the table below. Calculate the tank pressure in kPa. (1 L = 10^{-3} m^3)

	Element	Molar Mass (g/mol)
1	Helium	4.00
2	Neon	20.2
3	Argon	40.0
4	Krypton	83.8

p = _____kPa (Note the units!)

Question 2

0 / 10 pts

An ideal gas has an initial pressure of p_0 and volume of 1.7 m^3. It undergoes a process whereby the pressure changes to $(1.8)p_0$ while the Kelvin temperature increases by 67%. Calculate the final volume in meters cubed.

Question 3

0 / 10 pts

An ideal gas increases in temperature from 237°C to 480°C while the volume decreases by 47%. If the initial pressure is 74,643 Pa, what is the final pressure in kPa?

p_f = _____kPa (Note the units!)

Question 4

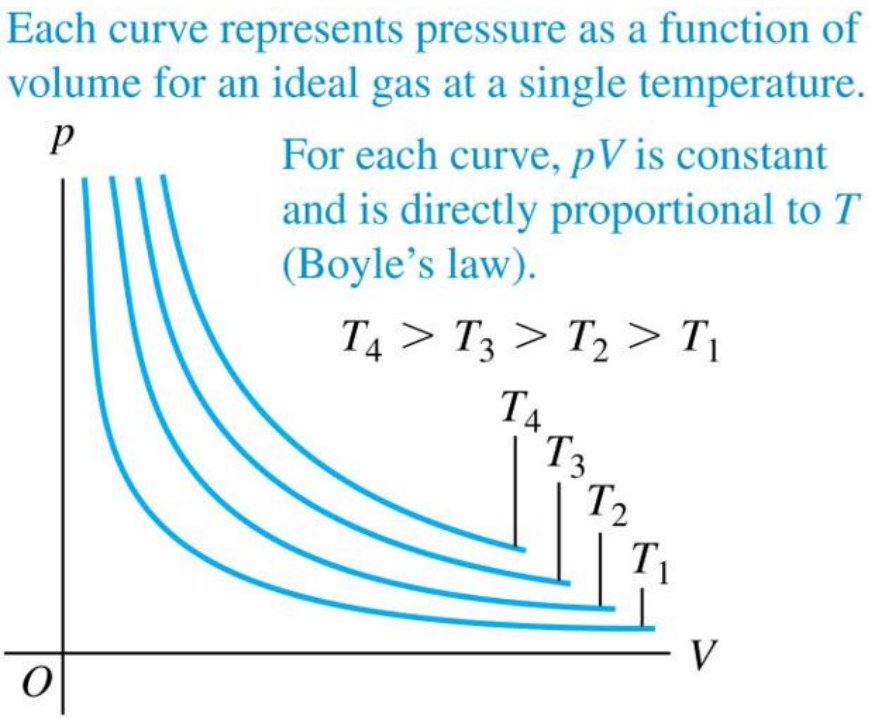
0 / 10 pts

A small bubble forms at a depth of 19 m in a freshwater lake. If the bubble's initial volume is 136 mm^3 and it expands without change in temperature as it rises, what is its volume in mm^3 just as it reaches the surface? Assume the pressure at the surface is 98,360 Pa and the water's density is 1000 kg/m^3.

V = _____ mm^3

pV-Diagrams (1 of 2)

- These show isotherms, or constant-temperature curves, for a constant amount of an ideal gas.



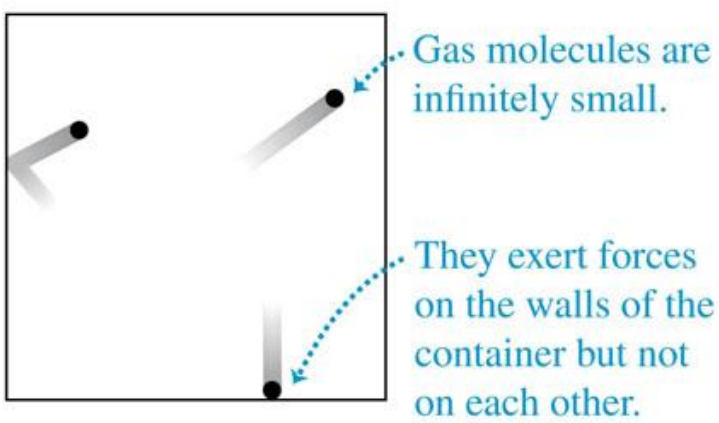
Introduction (2 of 2)

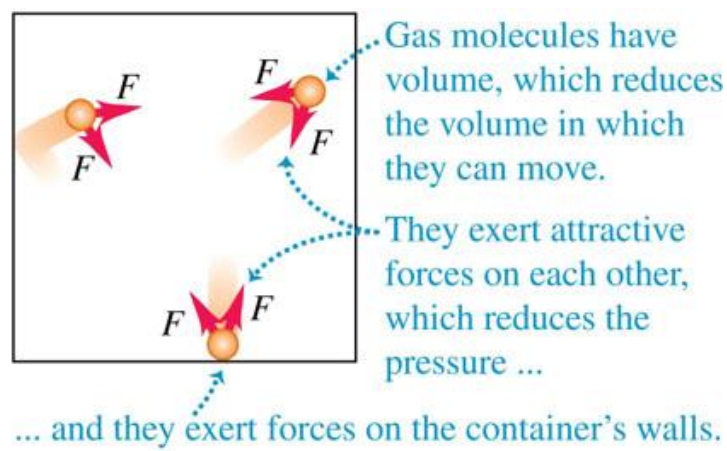
- The ideal-gas equation $pV = nRT$ gives a good description of the air inside an inflated vehicle tire, where the pressure is about 3 atmospheres and the temperature is much too high for nitrogen or oxygen to liquefy.
- As the tire warms (T increases), the volume V changes only slightly but the pressure p increases.

The Van Der Waals Equation

- The model used for the ideal-gas equation ignores the volumes of molecules and the attractive forces between them.
- The **van der Waals equation** is a more realistic model:

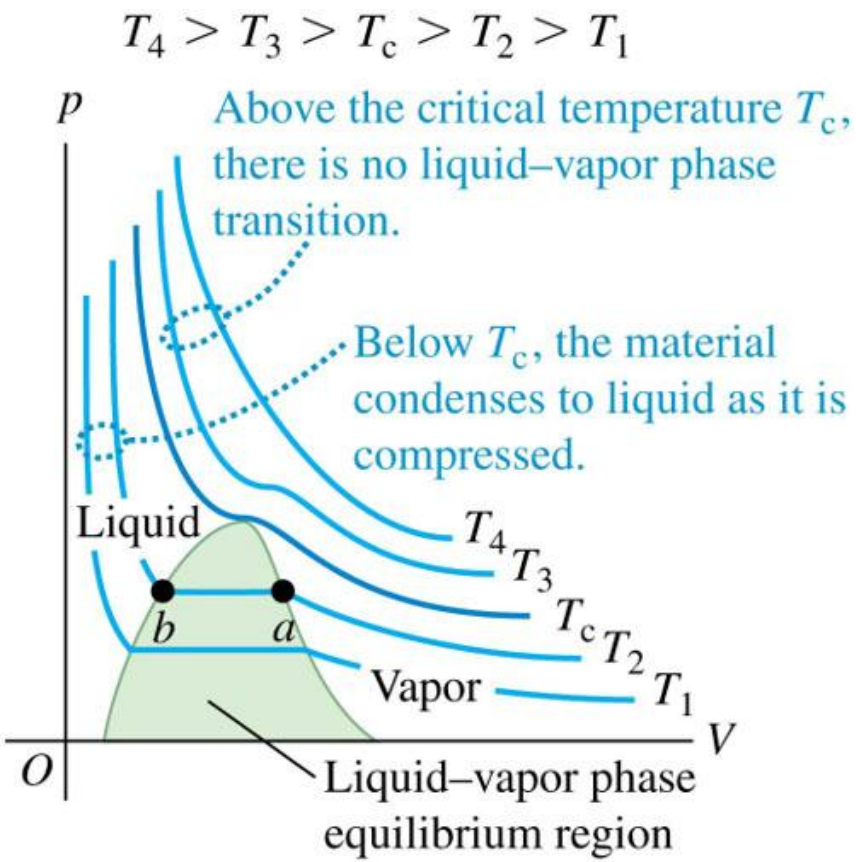
$$\left(p + \frac{an^2}{V^2}\right)(V - nb) = nRT$$





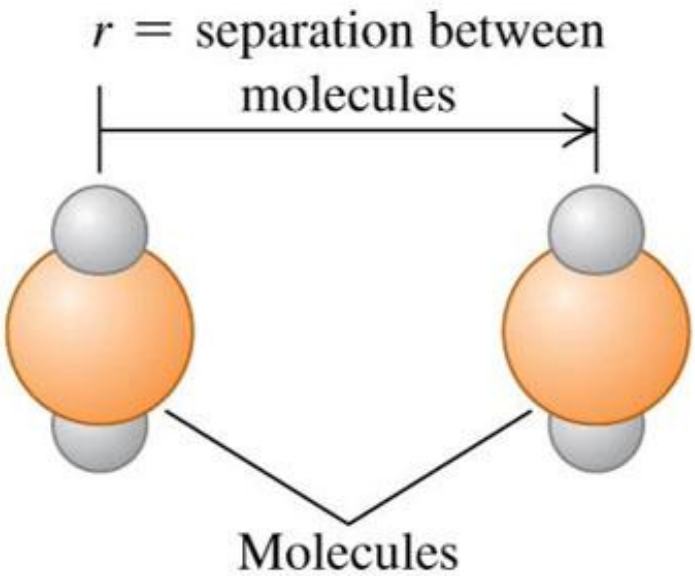
pV-Diagrams (2 of 2)

- A *pV*-diagram for a nonideal gas shows isotherms for temperatures above and below the critical temperature T_c.
- At still lower temperatures the material might undergo phase transitions from liquid to solid or from gas to solid.



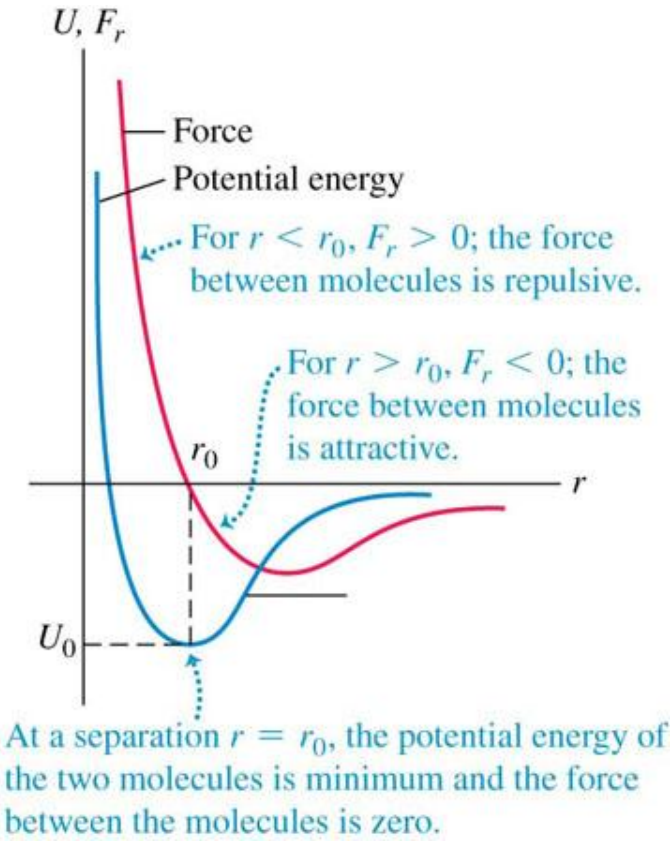
Molecules and Intermolecular Forces

- Any specific chemical compound is made up of identical **molecules**.
- In gases the molecules move nearly independently.
- The force between molecules in a gas varies with the distance *r* between molecules.
- When molecules are far apart, the intermolecular forces are very small and usually attractive.



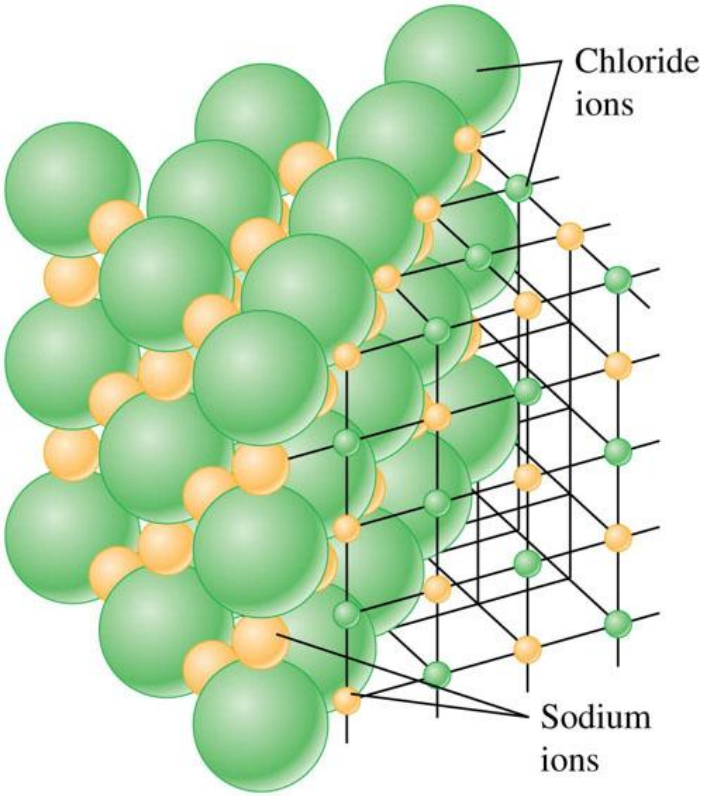
Molecular Properties of Matter (1 of 3)

- Figure 18.8 at the right shows how the force between molecules and their interaction potential energy depend on their separation *r*.
- Molecules in solids are essentially fixed in place, while those in liquids and gases have much more freedom to move.



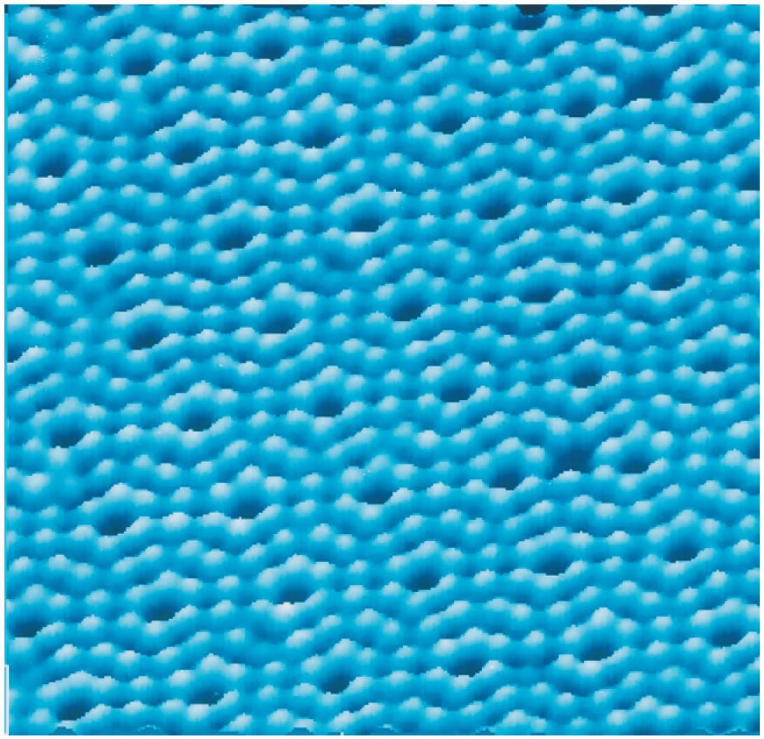
Molecular Properties of Matter (2 of 3)

- This is the schematic representation of the cubic crystal structure of sodium chloride (ordinary salt).



Molecular Properties of Matter (3 of 3)

- This is a scanning tunneling microscope image of the surface of a silicon crystal.
- The area shown is only 9.0 nm $(9.0 \times 10^{-9}\,\mathrm{m})$ across.
- Each blue “bead” is one silicon atom; these atoms are arranged in a (nearly) perfect array of hexagons.



Moles and Avogadro's Number

- One **mole** of a substance contains as many elementary entities (atoms or molecules) as there are atoms in 0.012 kg of carbon-12.
- One mole of a substance contains **Avogadro's number** N_A of molecules.
- $N_A = 6.022 \times 10^{23}$ molecules/mol
- The **molar mass *M*** is the mass of one mole.

Molar mass of a substance → $M = N_A m$ ← **Avogadro's number**; **Mass of a molecule** of substance

- When the molecule consists of a single atom, the term **atomic mass** is often used instead of molar mass.
- Video Tutor Solution: Example 18.5

Question 5

0 / 10 pts

Calculate the number of moles in 97 g of the molecule C_2H_5.

Question 6

0 / 10 pts

Calculate the mass of one molecule of C_2H_6OH in kg and supply the missing prefix below.

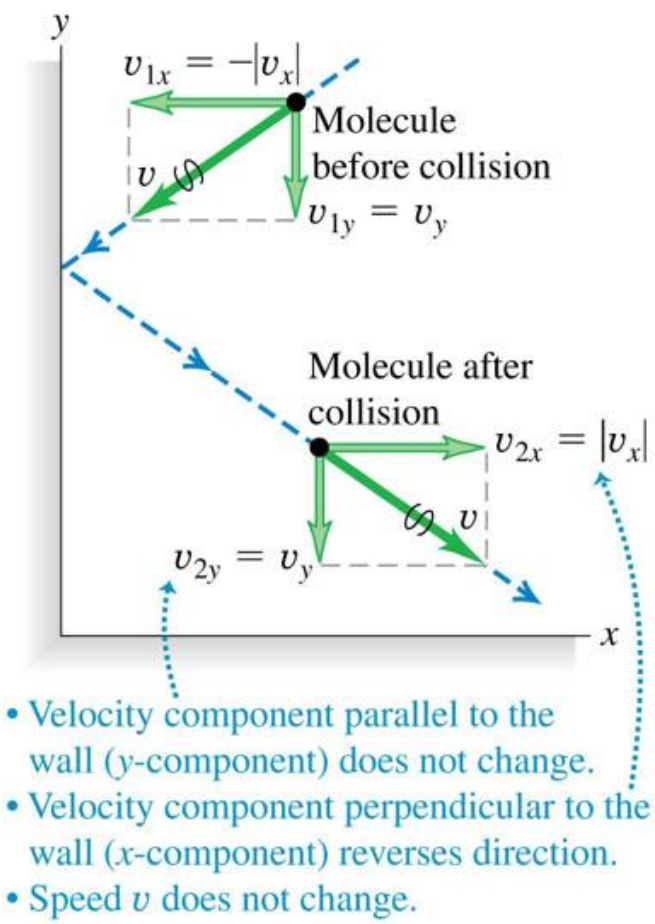
m = _____ $\times 10^{-27}$ kg

Kinetic-Molecular Model of an Ideal Gas

- The assumptions of the **kinetic-molecular model** are:
  1. A container contains a very large number of identical molecules.
  2. The molecules behave like point particles that are small compared to the size of the container and the average distance between molecules.
  3. The molecules are in constant motion and undergo perfectly elastic collisions.
  4. The container walls are perfectly rigid and do not move.

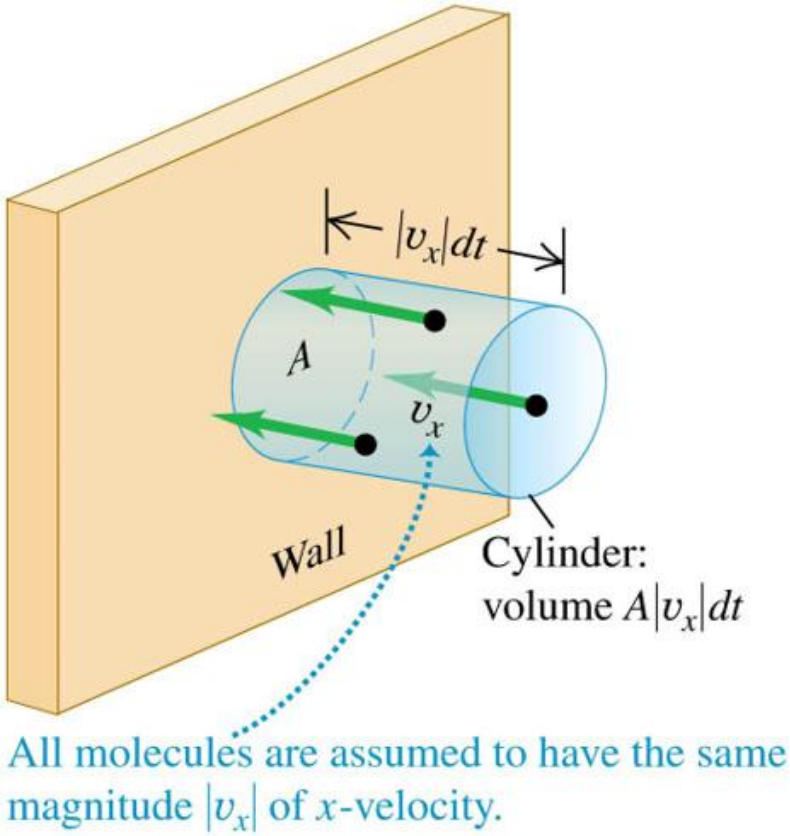
Collisions and Gas Pressure (1 of 2)

- During collisions the molecules exert **forces** on the walls of the container; this is the origin of the **pressure** that the gas exerts.
- In a typical collision (shown) the velocity component parallel to the wall is unchanged, and the component perpendicular to the wall reverses direction but does not change in magnitude.



Collisions and Gas Pressure (2 of 2)

- If a molecule is going to collide with a given wall area *A* during a small time interval *dt*, it must be within a distance $|v_x|dt$ from the wall (shown) and it must be headed toward the wall.
- So the number of molecules that collide with *A* during *dt* is equal to the number of molecules within the cylinder that have their *x*-velocity aimed toward the wall.



Pressure and Molecular Kinetic Energies

- The total random kinetic energy K_{tr} of translational motion of all the molecules in a gas is directly proportional to the absolute temperature T:

Average translational kinetic energy of an ideal gas

$$K_{tr} = \frac{3}{2}nRT$$

n: Number of moles of gas
T: Absolute temperature of gas
R: Gas constant

- This means that the average translational kinetic energy **per molecule** depends only on the temperature, not on the pressure, volume, or kind of molecule.

Average translational kinetic energy of a gas molecule:

$$\frac{1}{2}m\left(v^2\right)_{av.} = \frac{3}{2}kT$$

m: Mass of a molecule
$\left(v^2\right)_{av.}$: Average value of the square of molecular speeds
T: Absolute temperature of gas
k: Boltzmann constant

Question 7

0 / 10 pts

The average translational kinetic energy of a single atom is 9.2×10^{-21} J in a container of helium. Calculate the temperature of the gas as a whole.

Question 8

0 / 10 pts

Calculate the total translational kinetic energy in J of 6.1 moles of an ideal gas at a temperature of 90°C.

Molecular Speeds

- The root-mean-square speed of the molecules in a gas is:

Root-mean-square speed of a gas molecule

Boltzmann constant

Absolute temperature of gas

$$v_{\text{rms}} = \sqrt{(v^2)_{\text{av}}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$$

Average value of the square of molecular speeds

Mass of a molecule

Gas constant

Molar mass

- To compute the rms speed, we square each molecular speed, add, divide by the number of molecules, and take the square root; v_{rms} is the *root* of the **mean** of the **squares**.

- Video Tutor Solution: Example 18.7

Question 9

0 / 10 pts

Calculate the rms speed in m/s of molecular oxygen at a temperature of 17.3°C. (Molar mass of atomic oxygen is 16.0 g/mol.)

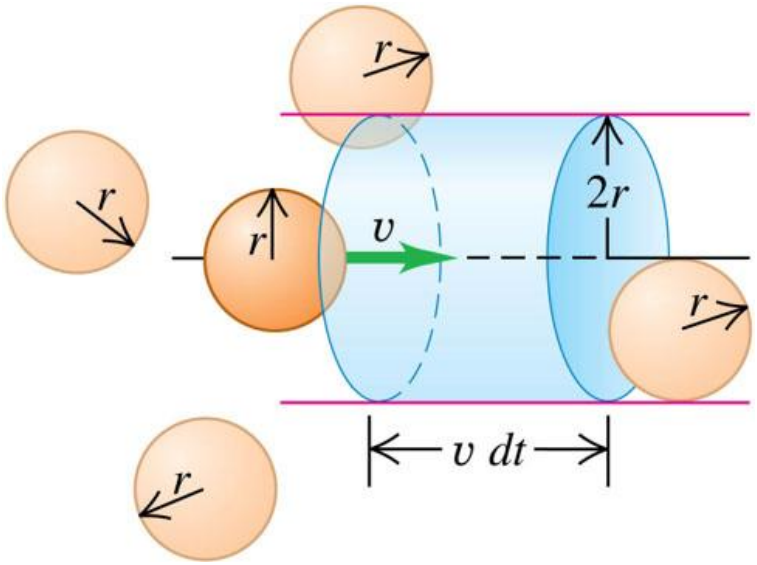
Question 10

0 / 10 pts

Suppose the root-mean-square speed of a gas is 557 m/s. If the molecular weight in grams per mole of the gas is 16 g/mol, calculate the temperature of the gas in Kelvin.

Collisions Between Molecules (1 of 2)

- We model molecules as rigid spheres of radius *r* as shown at the right.
- The **mean free path** of a molecule is the average distance it travels between collisions.
- The average time between collisions is the **mean free time**.
- In a time *dt* a molecule with radius *r* will collide with any other molecule within a cylindrical volume of radius 2*r* and length *v dt*.



Collisions Between Molecules (2 of 2)

- The average distance traveled between collisions is called the **mean free path**.
- In our model, this is just the molecule's speed *v* multiplied by mean free time:

Mean free path of a gas molecule

Speed of molecule

Volume of gas

$$\lambda = v t_{\text{mean}} = \frac{V}{4\pi\sqrt{2}r^2N}$$

Mean free time between collisions

Radius of a molecule

Number of molecules in gas

- The more molecules there are and the larger the molecule, the shorter the mean distance between collisions.
- Video Tutor Solution: Example 18.8

Heat Capacities of Gases

- The **degrees of freedom** are the number of velocity components needed to describe a molecule completely.
- A monatomic gas has three degrees of freedom and a diatomic gas has five; the **equipartition of energy principle** states that each degree of freedom has $\frac{1}{2}kT$ of kinetic energy associated with it.
- For an ideal monatomic gas:

Molar heat capacity at constant volume, ideal gas of point particles → $C_V = \frac{3}{2}R$ ← Gas constant

- For an ideal diatomic gas:

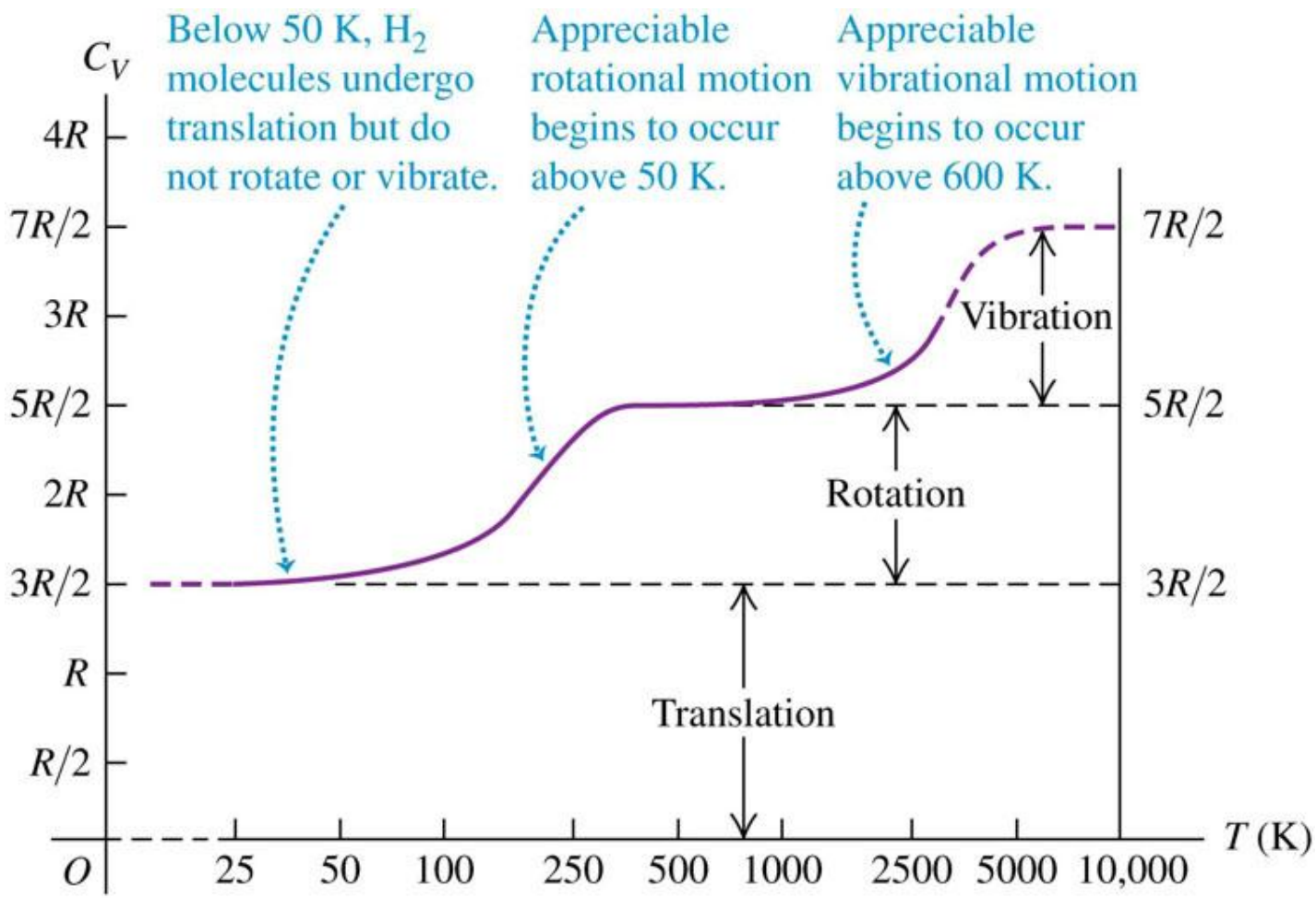
Molar heat capacity at constant volume, ideal diatomic gas → $C_V = \frac{5}{2}R$ ← Gas constant

Compare Theory with Experiment (1 of 2)

- Table 18.1 shows that the calculated values for C_V for monatomic gases and diatomic gases agree quite well with the measured values.

Type of Gas	Gas	C_v (j/mol·K)
Monatomic	He	12.47
	Ar	12.47
Diatomic	H_2	20.42
	N_2	20.76
	O_2	20.85
	CO	20.85
Polyatomic	CO_2	28.46
	SO_2	31.39
	H_2S	25.95

Experimental Values of CV for Hydrogen Gas (H_2)



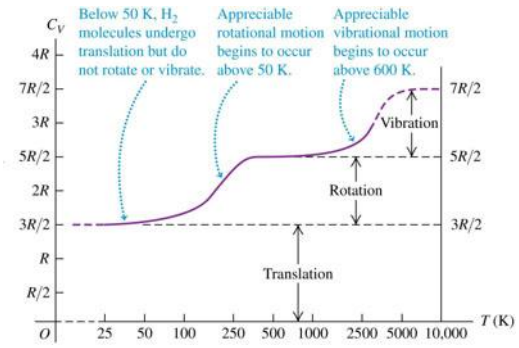
Heat Capacities of Solids

- Consider a crystal consisting of *N* identical atoms (**a monatomic solid**).
- Each atom is bound to an equilibrium position by interatomic forces.
- We can think of a crystal as an array of atoms connected by little springs.
- Each atom has an average kinetic energy $\frac{3}{2}kT$ and an average potential energy $\frac{3}{2}kT$, or an average total energy 3 kT per atom.
- The molar heat capacity of a crystal is:

Molar heat capacity of an ideal monatomic solid (rule of Dulong and Petit)

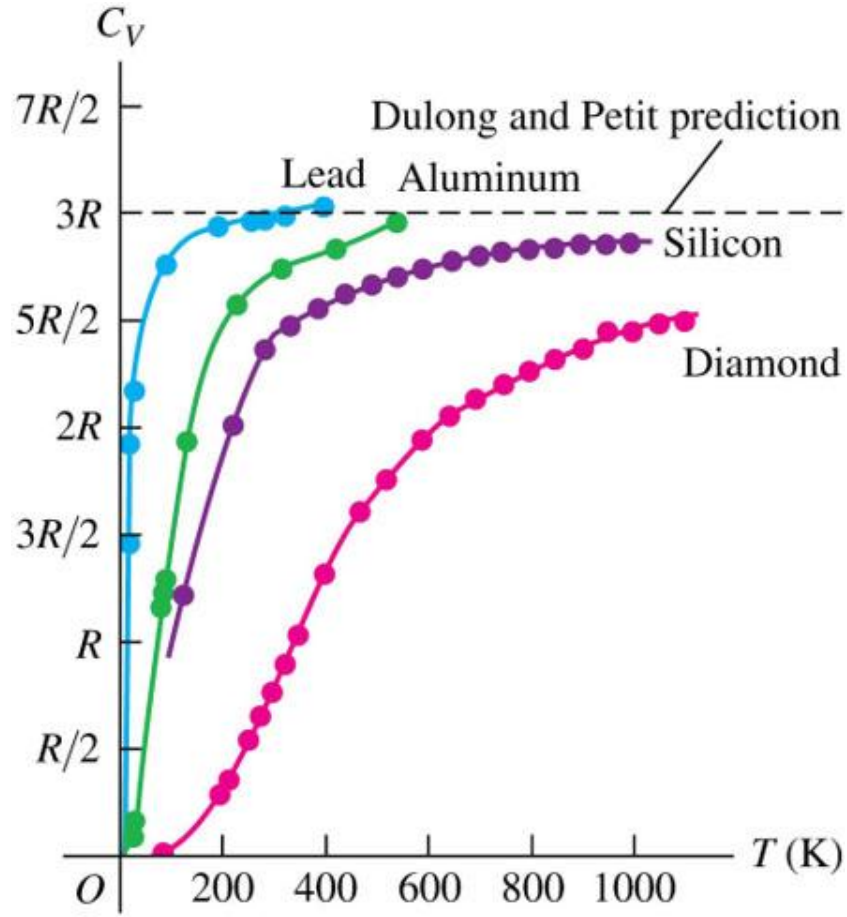
$$C_V = 3R$$

Gas constant



Compare Theory with Experiment (2 of 2)

- Experimental values of C_V for lead, aluminum, silicon, and diamond are given in the figure.
- At high temperatures, C_V for each solid approaches about $3R$, in agreement with the rule of Dulong and Petit.
- At low temperatures, C_V is much less than $3R$.



Question 12

0 / 10 pts

A monatomic solid has an atomic mass of 56.6 amu. Calculate the energy required to increase the temperature of 24 g of the substance by 15 C°.

Q = _____ J

Question 13

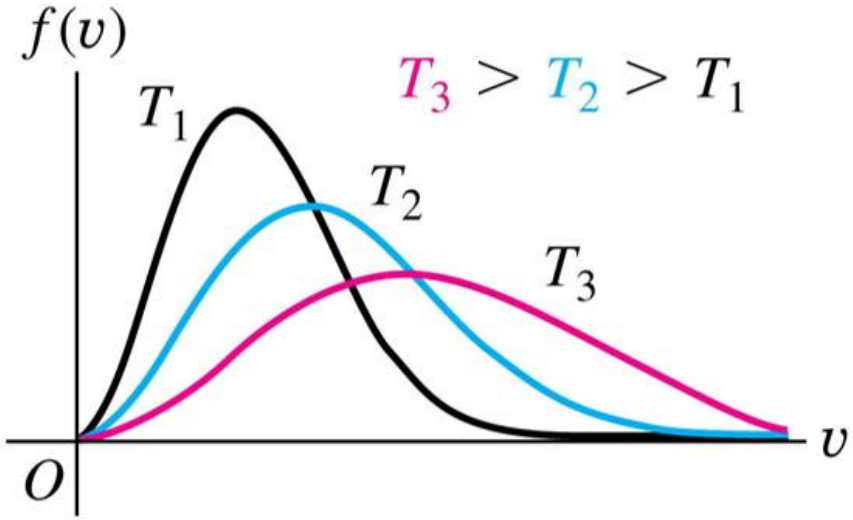
0 / 10 pts

Calculate the total internal energy in kJ of 7.8 mol of molecular hydrogen gas at a temperature of 412 K.

E = _____kJ

Molecular Speeds (1 of 3)

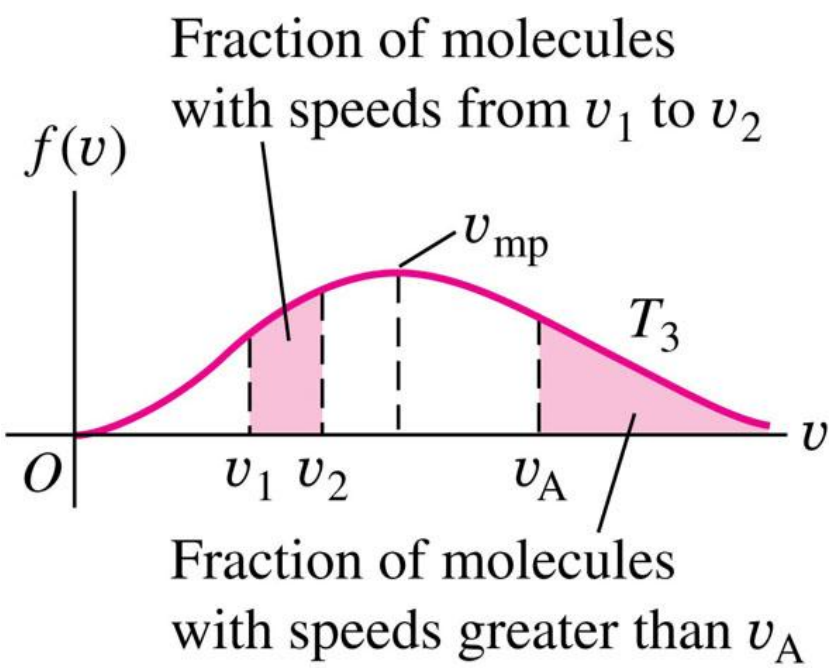
- The **Maxwell-Boltzmann distribution** *f* (*v*) gives the distribution of molecular speeds.



As temperature increases:
- the curve flattens.
- the maximum shifts to higher speeds.

Molecular Speeds (2 of 3)

- The most probable speed for a given temperature is at the peak of the curve.



Molecular Speeds (3 of 3)

- The function $f(v)$ describing the actual distribution of molecular speeds is called the **Maxwell–Boltzmann distribution**.
- It can be derived from statistical mechanics considerations, but that derivation is beyond our scope.
- Here is the result:

Maxwell–Boltzmann distribution function

$$f(v) = 4\pi\left(\frac{m}{2\pi kT}\right)^{3/2} v^2 e^{-mv^2/2kT}$$

Mass of a gas molecule

Molecular speed

Mass of a gas molecule

Molecular speed

Boltzmann constant

Boltzmann constant

Absolute temperature of gas

Question 11

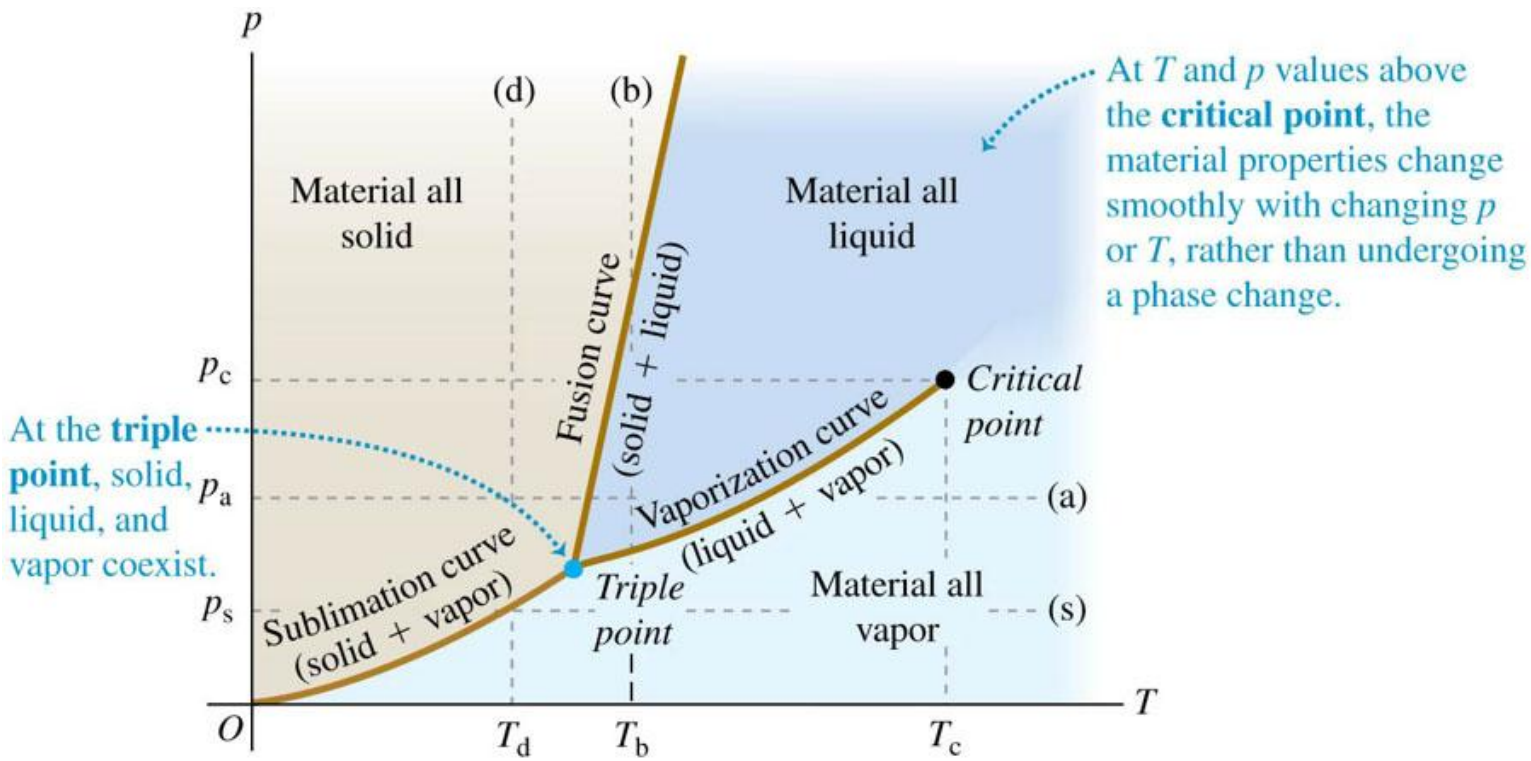
0 / 10 pts

Calculate the most probable speed (in m/s) of molecular oxygen at a temperature of 82.7 °C. (Molar mass of atomic oxygen is 16.0 g/mol.)

Phases of Matter (1 of 2)

- For an ideal gas we ignore the interactions between molecules.
- But those interactions are what makes matter condense into the liquid and solid phases under some conditions.
- Each phase is stable in only certain ranges of temperature and pressure.
- A transition from one phase to another ordinarily requires **phase equilibrium** between the two phases, and for a given pressure this occurs at only one specific temperature.
- We can represent these conditions on a graph with axes p and T, called a **phase diagram**. (See next slide.)
- Each point on the diagram represents a pair of values of p and T.

A Typical pT Phase Diagram

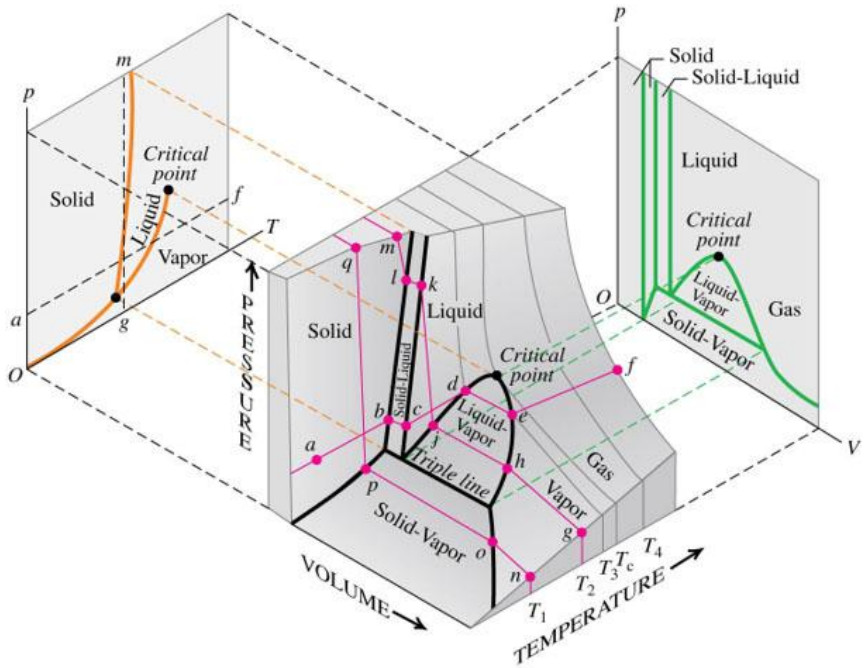


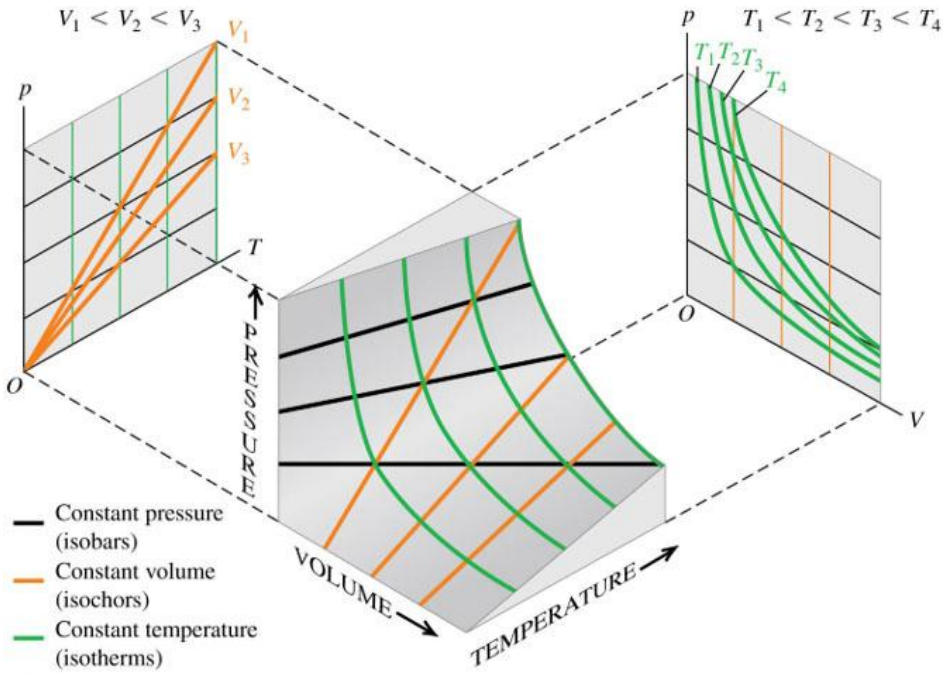
Phases of Matter (2 of 2)

- Atmospheric pressure on earth is higher than the triple-point pressure of water.
- Depending on the temperature, water can exist as a vapor (in the atmosphere), as a liquid (in the ocean), or as a solid (like the iceberg shown here).

pVT-Surface for a Substance That Expands on Melting

- A **pVT**-surface graphically represents the equation of state. Projections onto the *pT*- and *pV*-planes are shown.



pVT-Surface for an Ideal Gas

- The **pVT**-surface for an ideal gas is much simpler than the **pVT**-surface for a real substance, as seen in the previous slide.